

discrepancies are identified the differential between the identified and execution price becomes extremely important. Implementation shortfall needs to be minimized through, for instance, a highly optimized trading architecture (Hanif and Smith, 2012a).

Taking the example from Dacorogna et al. (2001), for a trader interested in yen, the interrelation between the three currencies (Japanese yen, Great British pound and the US dollar) is captured in the dynamics:

$$GBP/JPY_{bid} = \frac{USD/JPY_{bid}}{USD/GBP_{ask}}, \quad GBP/JPY_{ask} = \frac{USD/JPY_{ask}}{USD/GBP_{bid}}.$$

These dynamics represent the triangular relationship between the three currencies. If a direct market between them all exists and these dynamics are strongly deviated from, the deviation can be exploited through a set of riskless transactions.

A frequently employed mechanism for identifying triangular arbitrage opportunity is the rate product:

$$\gamma(t) = \prod_{i=1}^3 r_i(t) \quad (13.12)$$

where $r_i(t)$ is an exchange rate at time t (Aiba et al., 2002). An arbitrage opportunity is available if $\gamma > 1$ but, again crucially, this shall only be realized if the transaction is completed at the arbitrage decision prices. Continuing from the above, if you initially hold yen there are two unique rate products which can be calculated:

$$\gamma_1(t) = (JPY/GBP_{bid}(t)) \cdot (GBP/USD_{bid}(t)) \cdot \left(\frac{1}{JPY/USD_{ask}(t)} \right) \quad (13.13)$$

$$\gamma_2(t) = \left(\frac{1}{JPY/GBP_{ask}(t)} \right) \cdot \left(\frac{1}{GBP/USD_{ask}(t)} \right) \cdot (JPY/USD_{bid}(t)) \quad (13.14)$$

These two rate products represent the universe of arbitrage opportunities for this set of exchange rates (Daniel et al., 2009).

Suppose we observe the quotes $JPY/GBP_{bid} = 0.00761$, $GBP/USD_{bid} = 1.60085$ and $JPY/USD_{ask} = 0.01100$. We check the implicit cross rate by Equation 13.4.

$$JPY/USD_{ask}^{implied} = JPY/GBP_{bid} \cdot GBP/USD_{bid} \quad (13.15)$$

$$0.01208 = 0.00761 \cdot 1.60085 \quad (13.16)$$